SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4: *Design network topologies — star, mesh, bus, and hybrid — using networking devices, transmission media, and cabling standards.* ***(BL2, BL6)***

Assessment Tool Weightage: 30%

Annexure A

Industry Problem Solving Task

Code of Conduct:

I S. Karthikeyan / 192419048 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

CSA07 — COMPUTER NETWORKS

CO4 Assessment Tool 2 — Industry Problem Solving Task: Answers

CO4: Design network topologies — star, mesh, bus, and hybrid — using networking devices, transmission media, and cabling standards

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Scenario 1. 20-Person Startup — Star Topology Design

Problem Analysis

A 20-person startup needs a network that is easy for a small (possibly non-specialist) team to manage, cheap enough for a startup budget, and simple to troubleshoot when something goes wrong — without over-engineering for a scale the company doesn't have yet.

Proposed Solution — Star Topology

A star topology fits this brief precisely because it centralizes every connection through one manageable device, giving the small team a single place to monitor and control the whole network, at a cost proportionate to just 20 users.

- All workstations, printers, and VoIP phones connect individually to one central switch — nothing depends on any other device's cable to stay online.
- A Wi-Fi access point off the same switch covers laptops and mobile devices without extra cabling runs.
- The switch uplinks to a router/firewall that handles internet access, DHCP, and NAT for the whole office.

Technical Implementation — Bill of Materials

Component	Recommendation	Purpose
Core switch	24-port managed L2/L3 switch (e.g. Cisco Catalyst / Netgear ProSafe)	Central connection point; room to grow past 20 devices
Cabling	Cat6 UTP, point-to-point to each device	Gigabit speeds, reliable up to 100 m — plenty for one office floor
Router/firewall	Cisco RV series or pfSense appliance	Internet access, DHCP, NAT, and basic security
Wireless	1 Wi-Fi access point, switch-connected	Coverage for laptops/mobile without new cable runs

Testing & Validation Plan

- Certify every Cat6 run with a cable tester (wire-map, length, and performance) before going live.
- Fault-isolation test: unplug one workstation's cable and confirm the other 19 stay fully connected.
- Ping/latency check between the switch and each device, plus a VoIP call-quality test on the phones.
- Wi-Fi coverage walk-test across the office to confirm no dead zones.

Documentation

A simple network diagram (switch, router, AP, and device count per port) plus an IP-addressing sheet is enough documentation at this scale — kept in a shared drive so any team member can hand it to an IT contractor later.

Why This Works

The key advantage is fault isolation: a single failed cable takes down only that one node, leaving all other 19 employees unaffected, and centralized cabling makes future moves/adds/changes simple as the startup grows.

Scenario 2. Bank Network — Mesh Topology for Near-Zero Downtime

Problem Analysis

A bank's core requirement is near-zero downtime — any outage risks financial transaction failures and breaches regulatory uptime obligations, so the design must eliminate single points of failure at every critical layer, not just add convenience redundancy.

Proposed Solution — Partial/Full Mesh at the Core

- Core routers, core switches, and firewalls connect directly to each other (full mesh at the core), so no single link or device failure isolates any part of the network.
- Distribution and access layers use redundant uplinks to at least two core switches rather than a single one.
- Dual enterprise-grade routers (e.g. Cisco ASR / Juniper MX) provide BGP-based redundant internet connectivity.
- All core links run over fiber (single-mode between buildings, multi-mode OM4 within the data center) for EMI immunity, high bandwidth, and the low latency financial transactions require.

Technical Implementation — Bill of Materials

Component	Recommendation	Purpose
Core layer	Full mesh of core switches + firewalls	No single link/device failure can isolate the network
Routing	Dual Cisco ASR / Juniper MX routers, BGP	Redundant, automatically-failing-over internet connectivity
Core links	Single-mode fiber (inter-building), OM4 multi-mode (in-DC)	EMI immunity, high bandwidth, low latency
Power & storage	Redundant PSUs, UPS, RAID arrays	Protects against power loss and disk failure

Testing & Validation Plan

- Simulated link-failure test: disconnect a core fiber link and measure OSPF/BGP reroute convergence time (should be in the millisecond-to-low-second range).
- Load testing at simulated peak transaction volume to confirm throughput holds under real banking-hours demand.
- UPS/power-failover drill and RAID rebuild test to confirm data integrity survives a hardware fault.
- Security/penetration testing on firewalls and routers to satisfy regulatory compliance requirements.

Documentation

Given the regulatory context, this design needs formal documentation: a full logical/physical topology diagram, a failover run-book, and recorded test results from every drill above, retained for audit purposes.

Why This Works

When a link fails in a mesh, traffic reroutes automatically through alternative paths in milliseconds via OSPF/BGP — satisfying the bank's regulatory uptime and data-integrity requirements in a way a single-path topology never could.

Scenario 3. Small Classroom Lab — Bus Topology on a Tight Budget

Problem Analysis

A small, supervised classroom lab with light usage (browsing, file sharing) has one dominant constraint — budget — and does not need the fault tolerance or performance headroom a bank or startup would require.

Proposed Solution — Bus Topology

- Classic option: a single RG-58 (50-ohm) coaxial backbone cable, with each computer T-connected via a BNC connector, and 50-ohm terminators at both ends to prevent signal reflection.
- Modern low-cost variant: a single inexpensive 8- or 16-port unmanaged switch with Cat5e UTP patch cables to each workstation — keeping bus-like simplicity and cost with easier cabling.
- No dedicated server needed — a teacher's PC shares resources using built-in Windows/Linux file sharing.

Technical Implementation — Bill of Materials

Component	Recommendation	Purpose
Backbone	RG-58 coax (classic) or 1 unmanaged switch (modern)	Shared communication line for all lab PCs
Connectors	BNC + T-connectors (coax variant)	Tap each PC onto the shared backbone
Terminators	50-ohm terminators at both cable ends	Absorb signal, prevent reflection/echo on the line
Cabling (modern variant)	Cat5e UTP patch cables	Connect each PC to the low-cost unmanaged switch

Testing & Validation Plan

- Continuity test on the backbone cable and terminators before connecting any PCs.
- Simultaneous-use test — have several PCs transfer files at once and observe collision/slowdown behaviour under realistic classroom load.
- Basic connectivity test: verify every PC can reach the teacher's shared folder.

Documentation

A one-page cable run sheet (which PC sits at which tap position) is sufficient documentation here — the network is small and static enough that anything more elaborate would be wasted effort.

Why This Works

The network is susceptible to collisions and a single cable break disrupts every connected machine, but for a supervised classroom with light, non-critical usage, that trade-off is entirely acceptable in exchange for the lowest possible hardware cost.

Scenario 4. Large University Campus — Hybrid Topology

Problem Analysis

A campus with thousands of users spread across lecture halls, labs, libraries, dormitories, and administration buildings cannot be served well by any single topology — it needs backbone-level redundancy, building-level manageability, and lab-level cost-efficiency all at once.

Proposed Solution — Hierarchical Hybrid (Core / Distribution / Access)

- Campus backbone: high-speed fiber (single-mode, 10/40 Gbps) connects core routers and multilayer switches in a ring or partial-mesh arrangement, so no single link failure isolates an entire building.
- Each building has its own distribution switch, connected to the campus core via dual fiber uplinks for redundancy.
- Within each building, a star topology branches from the distribution switch down to floor-level access switches, which connect individual devices via Cat6.
- Wireless access points connect via PoE to access switches, removing the need for separate power cabling.
- Individual labs use bus/tree structures internally where cost-efficiency matters more than redundancy.

Technical Implementation — Bill of Materials

Layer	Design	Purpose
Core	Fiber ring/partial mesh, 10-40 Gbps single-mode	Backbone redundancy — no building isolated by one link failure
Distribution	1 switch per building, dual fiber uplinks to core	Redundant building-to-backbone connectivity
Access	Star from distribution to floor switches, Cat6 to devices	Simple, manageable in-building connectivity
Wireless	PoE access points off access switches	Coverage without separate power runs

Testing & Validation Plan

- Backbone failover test: simulate a single fiber-ring link cut and confirm automatic rerouting with no building losing connectivity.
- Inter-building latency and throughput testing during peak class-change hours.
- PoE power-budget verification to ensure the access switches can support every connected access point.

Documentation

This scale needs a formal three-tier network diagram (core/distribution/access), a building-by-building IP addressing/VLAN plan, and a change-log process, since new buildings will be added to the design over time.

Why This Works

This hierarchical core-distribution-access model is the industry standard for enterprise/campus networks precisely because adding a new building simply means installing a new distribution switch and running fiber to the core — with no disruption to the existing infrastructure.

Scenario 5. Multi-Department Company — Structured, VLAN-Segmented Network

Problem Analysis

A company with several departments (HR, Finance, Sales, IT) needs devices within each department to communicate freely, but also needs department-to-department traffic controlled for both security and broadcast-traffic efficiency — a flat, unsegmented network satisfies neither requirement well.

Proposed Solution — Layer 2 Access + Layer 3 Core with VLANs

- Each department connects through its own dedicated Layer 2 access switch, letting devices within that department (PCs, printers) communicate directly.
- All access switches connect up to a central Layer 3 switch/router, which performs inter-VLAN routing to allow controlled communication between departments.
- VLANs logically separate each department's traffic — improving security and cutting down unnecessary broadcast traffic between unrelated departments.
- A router with firewall functionality connects the internal network to the internet, controlling and securing external access.
- Redundant links/backup paths and a structured IP addressing/subnetting scheme support reliability and manageability as the company grows.

Technical Implementation — Bill of Materials

Component	Recommendation	Purpose
Access layer	1 Layer 2 switch per department	Local intra-department device connectivity

Component	Recommendation	Purpose
Core layer	Layer 3 switch/router with inter-VLAN routing	Controlled communication between departments
Segmentation	One VLAN per department	Security isolation and reduced broadcast traffic
Perimeter	Firewall-enabled router	Secure internet access for the whole company

Testing & Validation Plan

- Inter-VLAN routing test — ping between two departments' devices to confirm the Layer 3 switch routes traffic correctly where permitted.
- Broadcast-isolation test — confirm a broadcast in one VLAN does not appear in another department's VLAN.
- Firewall rule verification for internet access and any inter-department restrictions.
- Redundant-link failover test on the backup paths between access and core switches.

Documentation

A VLAN and IP-subnetting plan (department → VLAN ID → subnet range), plus a topology diagram showing access-to-core connections, gives IT staff everything needed to manage or extend the network later.

Why This Works

This hierarchical, VLAN-segmented design improves performance, strengthens security through traffic isolation, and allows easy expansion — a new department simply gets its own access switch and VLAN without disturbing any existing department's network.